INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS553)

LAB 13

Assignment 13: Recursion on Arrays

Assignment aim: Process array elements recursively.

Experiment No. 13.1 — Recursive sum of array

1. Experiment Title: Recursive sum of array

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Recursive array methods move an index toward a stopping condition to sum, find a maximum or print in reverse.

5. Algorithm / Procedure:
If the index reaches array length, return zero; otherwise add the current element to the recursive sum of the remaining elements.
Then compile and execute the Java program.

6. Program:

```
public class Task_01_Recursive_sum_of_array {
    static int sum(int[] a, int i) {
        if (i == a.length) return 0;
        return a[i] + sum(a, i + 1);
    }
    public static void main(String[] args) {
        int[] a = {3, 6, 9, 12, 15};
        System.out.println(sum(a, 0));
    }
}
```

7. Input: Fixed input values are declared in the program: array {3, 6, 9, 12, 15}.

8. Output:
45

Time / Space Complexity: O(n) time; O(n) call-stack space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is the stopping condition?
- Why does the index increase?
- What is the sum of the given array?

Experiment No. 13.2 — Find array maximum recursively

1. Experiment Title: Find array maximum recursively

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Recursive array methods move an index toward a stopping condition to sum, find a maximum or print in reverse.

5. Algorithm / Procedure:
When the last index is reached, return that element; otherwise compare the current value with the maximum of the rest.
Then compile and execute the Java program.

6. Program:

```
public class Task_02_Find_array_maximum_recursively{
    static int max(int[] a, int i) {
        if (i == a.length - 1) return a[i];
        return Math.max(a[i], max(a, i + 1));
    }
    public static void main(String[] args) {
        int[] a = {3, 6, 9, 12, 15};
        System.out.println(max(a, 0));
    }
}
```

7. Input: Fixed input values are declared in the program: array {3, 6, 9, 12, 15}.

8. Output:
15

Time / Space Complexity: O(n) time; O(n) call-stack space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- Why is the last element the base case?
- What does Math.max return?
- What is the maximum value here?

Experiment No. 13.3 — Print array in reverse recursively

1. Experiment Title: Print array in reverse recursively

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Recursive array methods move an index toward a stopping condition to sum, find a maximum or print in reverse.

5. Algorithm / Procedure:
Start at the last index; print the current value; recursively move to the previous index; stop when index is negative.
Then compile and execute the Java program.

6. Program:

```
public class Task_03_Print_array_in_reverse_recursively{
    static void printReverse(int[] a, int i) {
        if (i < 0) return;
        System.out.println(a[i]);
        printReverse(a, i - 1);
    }
    public static void main(String[] args) {
        int[] a = {3, 6, 9, 12, 15};
        printReverse(a, a.length - 1);
    }
}
```

7. Input: Fixed input values are declared in the program: array {3, 6, 9, 12, 15}.

8. Output:
15
12
9
6
3

Time / Space Complexity: O(n) time; O(n) call-stack space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- Why start from the last index?
- What is the base case?
- How does the index change on each call?